

Game Engine Concepts

Fachhochschule Oberösterreich

Human Interface Devices



Bad Game Interface Aspects:

Half-Life 2: Important tasks explained only once (via audio)



Bad Game Interface Aspects:

Skyrim: Ambiguous bag – what am I wearing, and how to compare items?

ALL

APPAREL

POTIONS

BOOKS

MISC

◇ College Robes

◇ Gloves

Lockpick (15)

◇ Mage Hood

Potion of Minor Healing (3)

Potion of Minor Stamina (5)

The Book of the Dragonborn



COLLEGE ROBES

ARMOR 0

WEIGHT 1

VALUE 10

[E] Equip [R] Drop [F] Favorite

Armor Rating 0 Carry Weight 11/300 Gold 34

Bad Game Interface Aspects:
Fallout 4: Why so small?



[(A) TOGGLE ACTIVE (X) SHOW ON MAP (D) NAVIGATE OBJECTIVES (Y) PERK CHART]

Bad Game Interface Aspects:
What the hell?





Bad Game Interface Aspects:

GTA San Andreas: Why do NPCs have different walking speeds than I have?



Bad Game Interface Aspects:

Resident Evil 2: Who can move correctly with these ill-placed cameras?

Human Interface Devices

Bad Examples

→ NES Classic Controller: Ergonomics, which hands would fit this?



Human Interface Devices

The Importance of Human-Machine Interfaces

“Better” interfaces guide the evolution of computing

Human Interface Devices

Examples

➔ From **command-line** interfaces...

```

40 GETA$: IFA$ < CHR$(13) THEN 40
42 PL=PL+1: IF PL>PY THEN PL=1
45 B$=""
    " : V$=""
50 TS=TI: TX=TS+6000: PC=1: PK=33268: D=1:
EM ** DRAW BORDER *
60 PRINT "[B$] PLAYER"PL" - "LE
T$(PN$(PL),10)SPC(12);
70 FOR I=1 TO 22: PRINT "[B$]";
75 NEXT: PRINT "[R] PRESS ! TO QUIT --
IFFICULTY ="(20-N%(PL))/2"[B$]"LEFT$(B$,4
80 T=INT(RND(1)*900+32768): IF (33248<T
ND(T<33288) THEN 80
82 TD=1: NN=N%(PL)
85 IF PEEK(T)<>32 THEN 80
90 POKE T,87: REM *** BEGIN ***
99 REM
100 FOR II=1 TO NN: IF PEEK(PE)=81 THEN POKE
BREAK
READY.

```

Human Interface Devices

Examples

→ From **command-line** interfaces...

→ ...to the “**Desktop Metaphor**”



Human Interface Devices

Examples

→ From **Personal Assistants...**



Human Interface Devices

Examples

→ From **Personal Assistants...**

→ To the **Smartphone**



Human Interface Devices

Examples

→ From messengers, blogs, and forums...

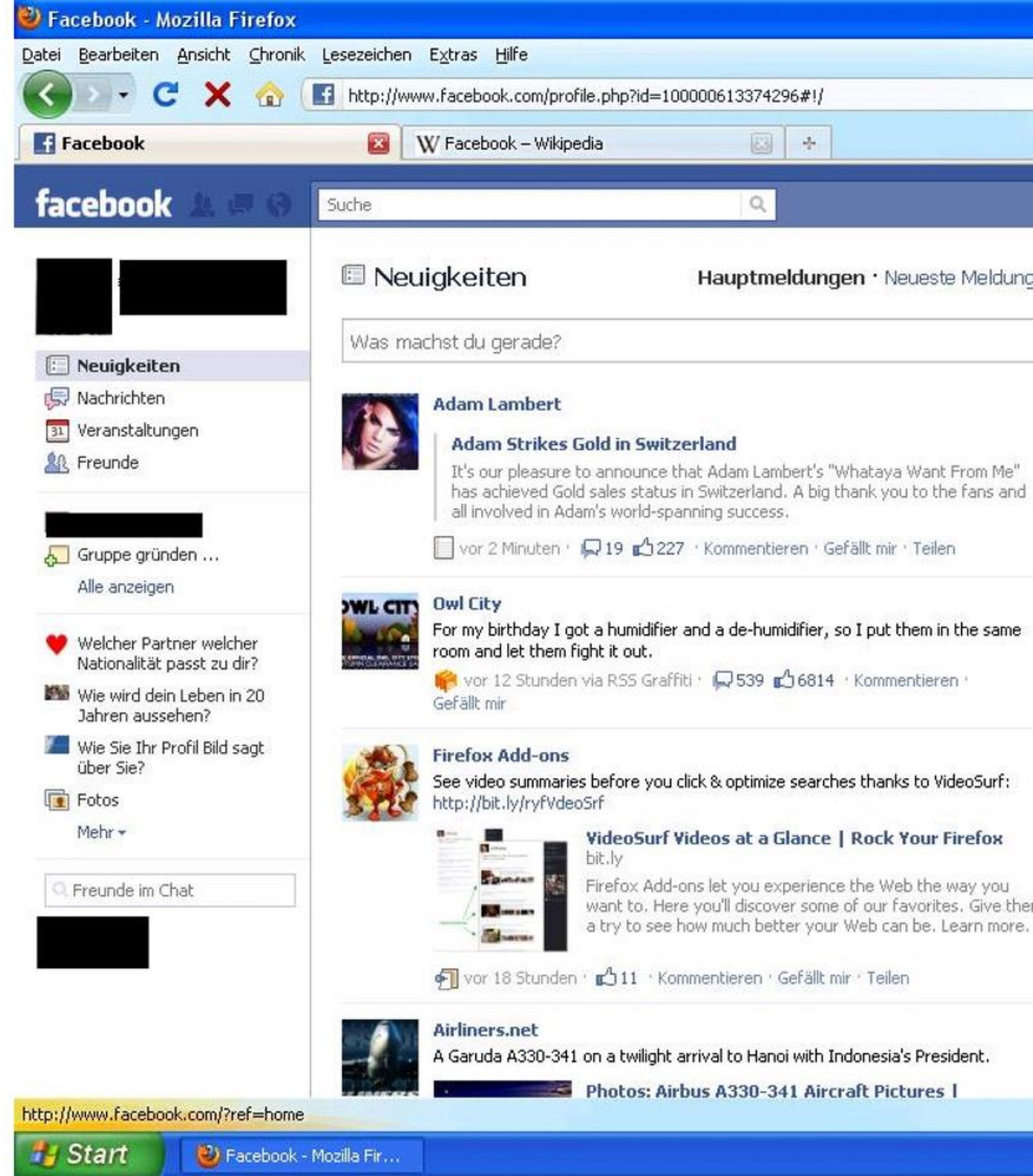


Human Interface Devices

Examples

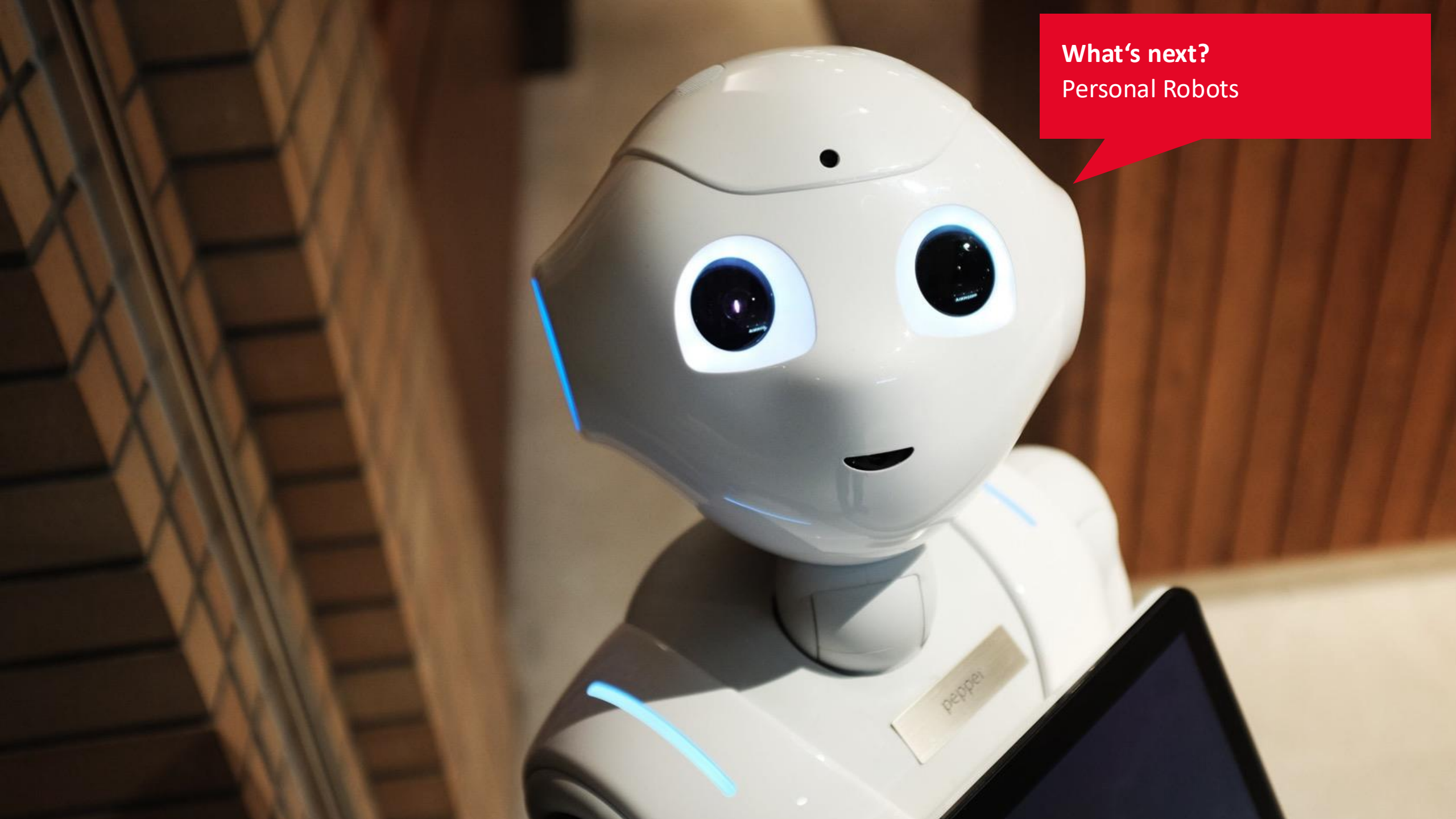
→ From messengers, blogs, and forums...

→ To Social Media





What's next?
Virtual Reality

A close-up, low-angle shot of the Pepper robot, a white, humanoid robot with large, expressive blue eyes and a small black dot for a nose. It is holding a tablet in its right hand. The background is a blurred wooden wall. A red speech bubble is in the top right corner.

What's next?
Personal Robots

What's next?
Automated Vehicles





Human Interface Devices

User Interface (UI)

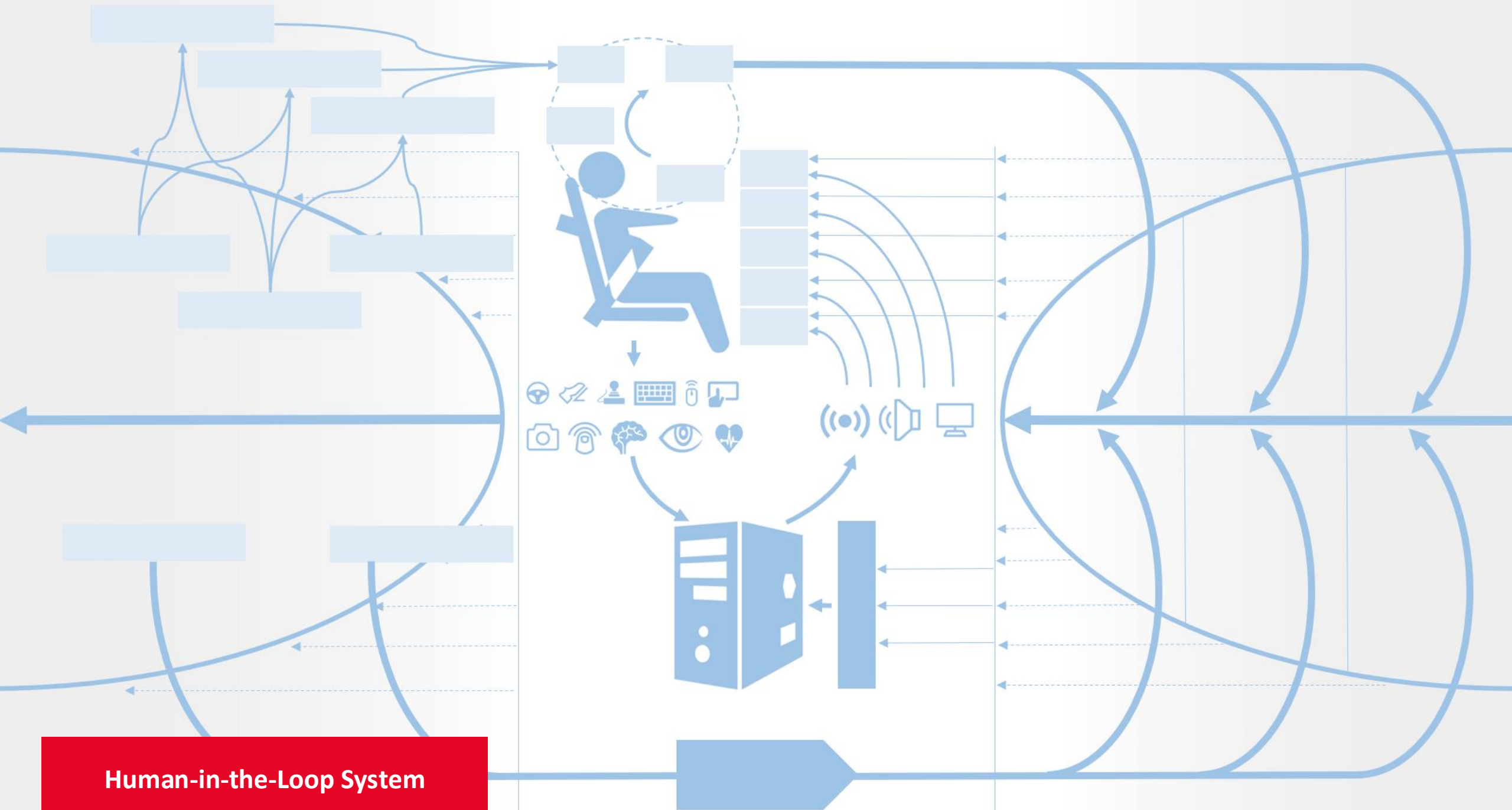
The interface between the user and the information has become the most critical issue for creating effective, efficient and pleasant systems

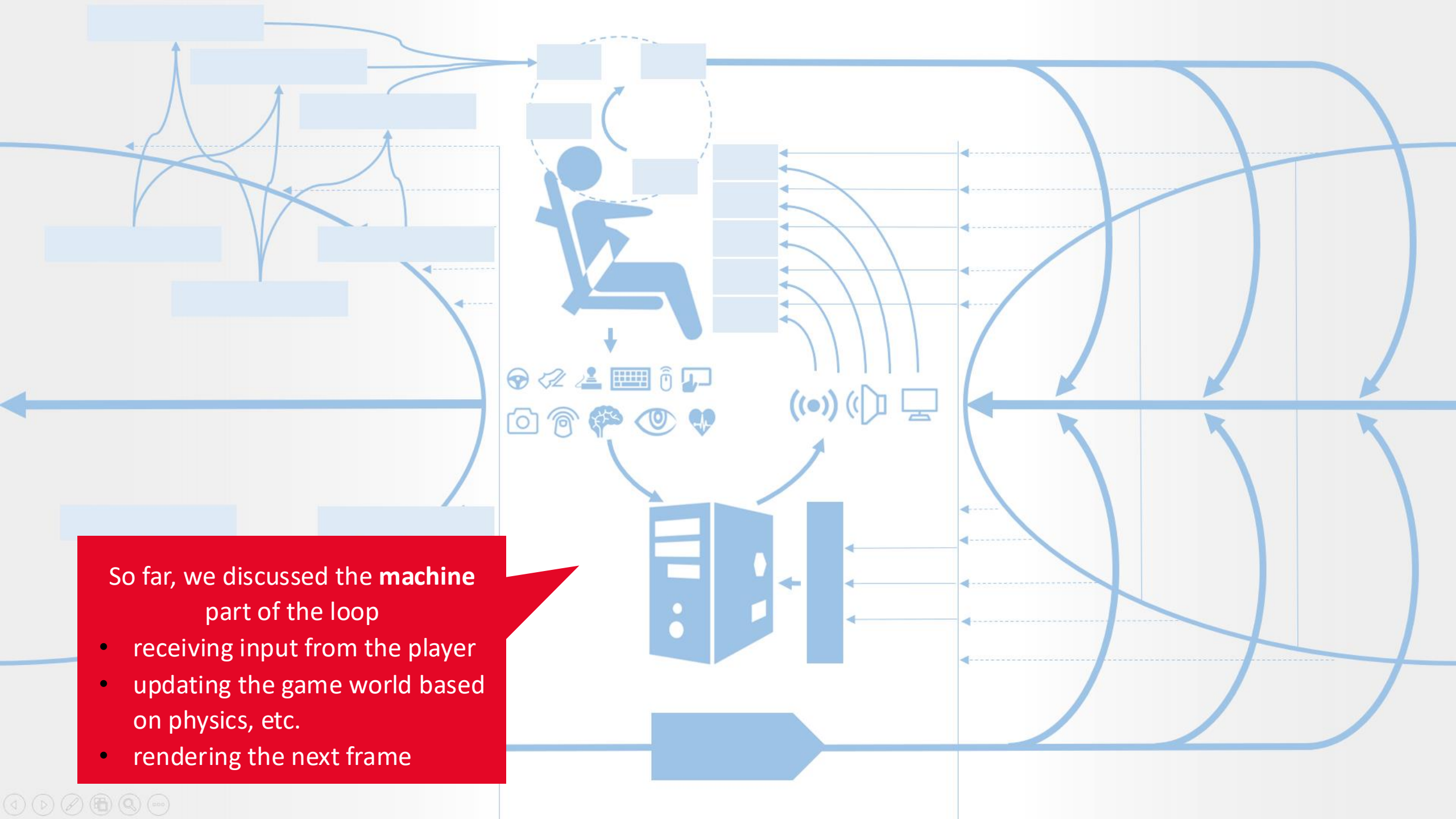


Interaction Design:

Allocating functions to humans and/or technology
Involves sequence/structure of touchpoints of space and time

Benyon, D. (2019). Designing User Experience: a guide to HCI, UX and interaction design. Pearson UK

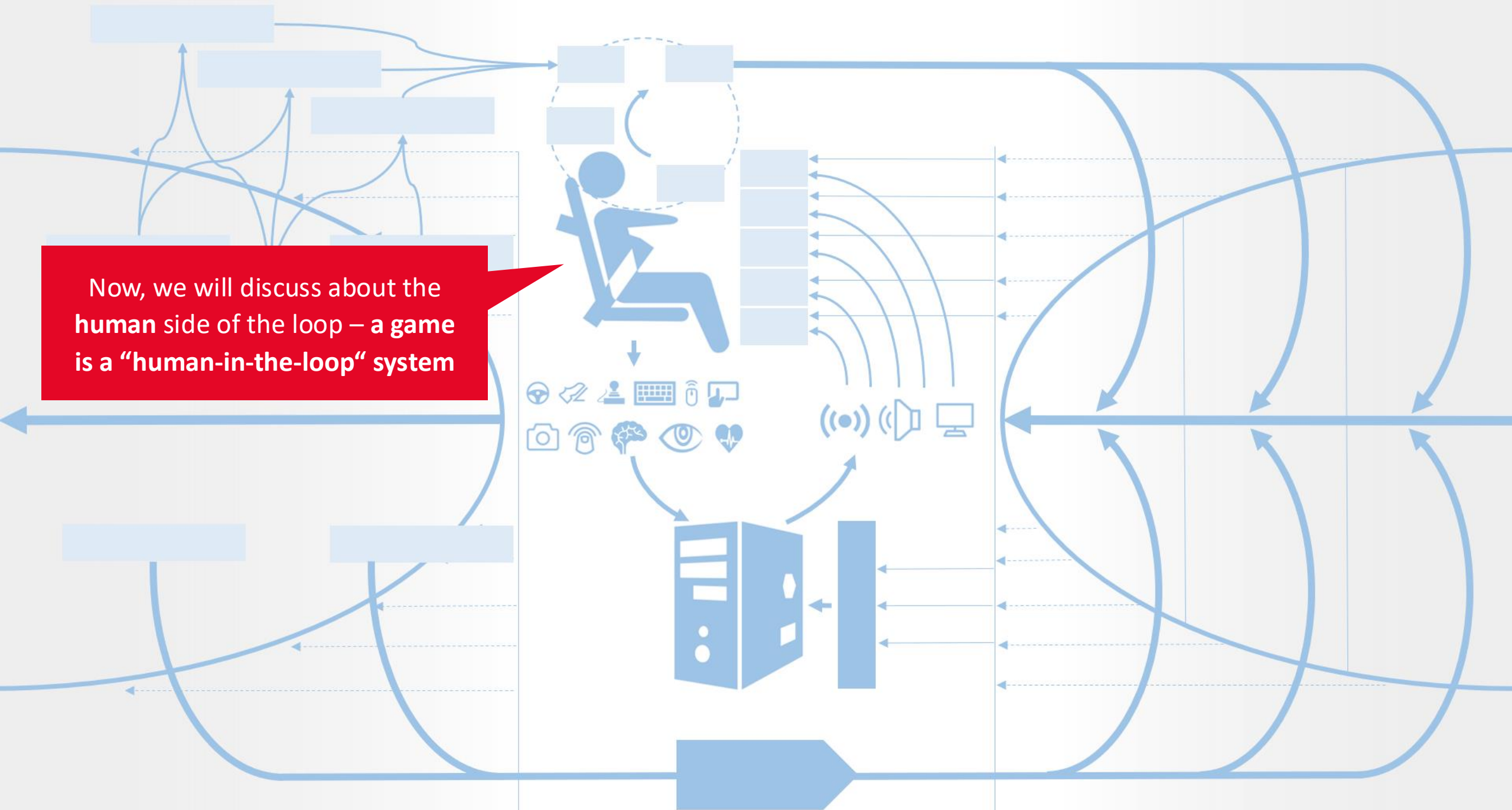




So far, we discussed the **machine** part of the loop

- receiving input from the player
- updating the game world based on physics, etc.
- rendering the next frame

Now, we will discuss about the human side of the loop – a game is a “human-in-the-loop” system



Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario



Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario
- 1) Output provided by the machine

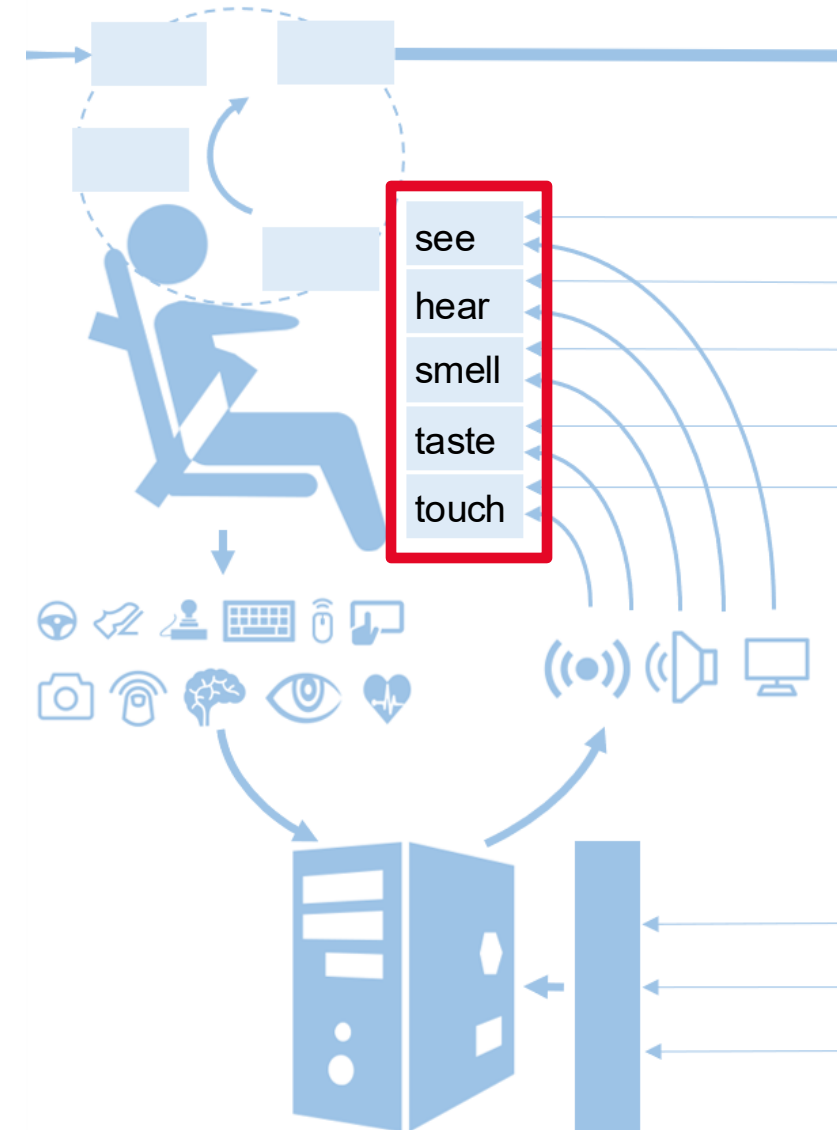


Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario
- 1) **Output** provided by the machine
- 2) Human **perception**

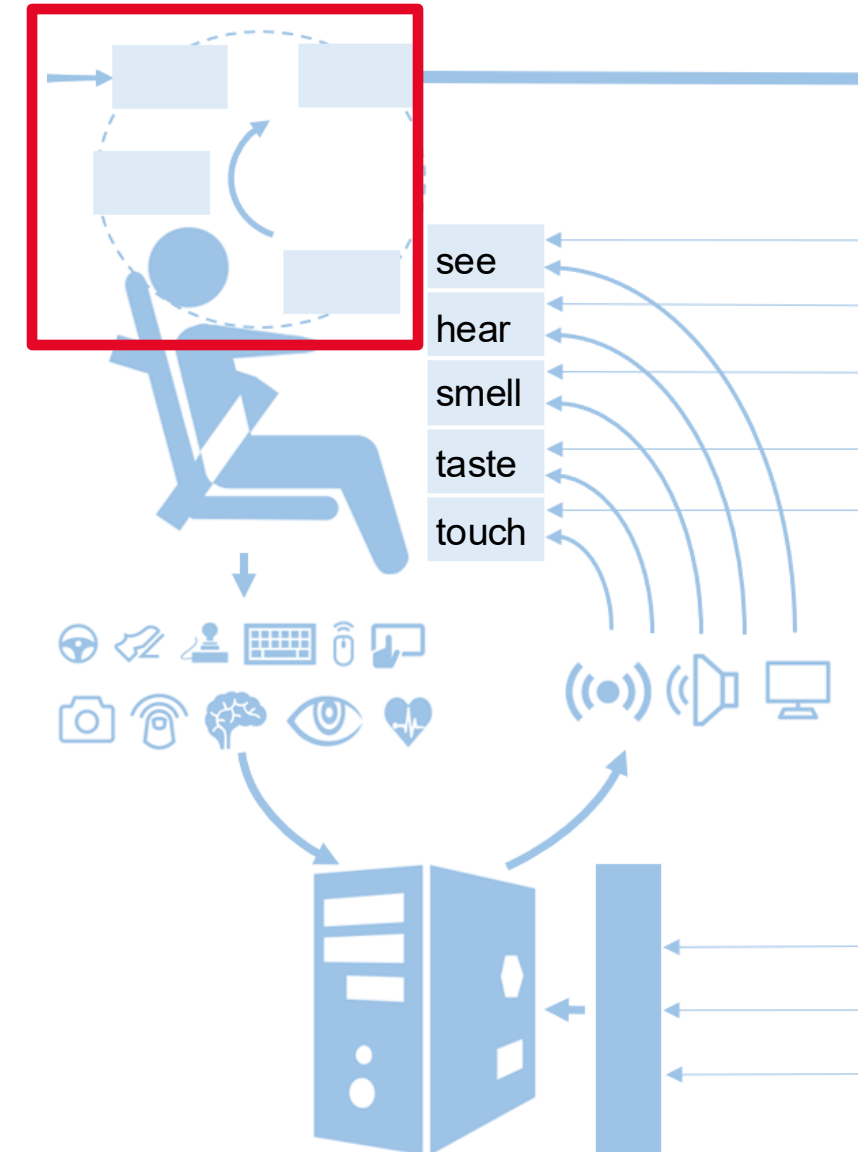


Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario
- 1) **Output** provided by the machine
- 2) Human **perception**
- 3) Human processing / decision making

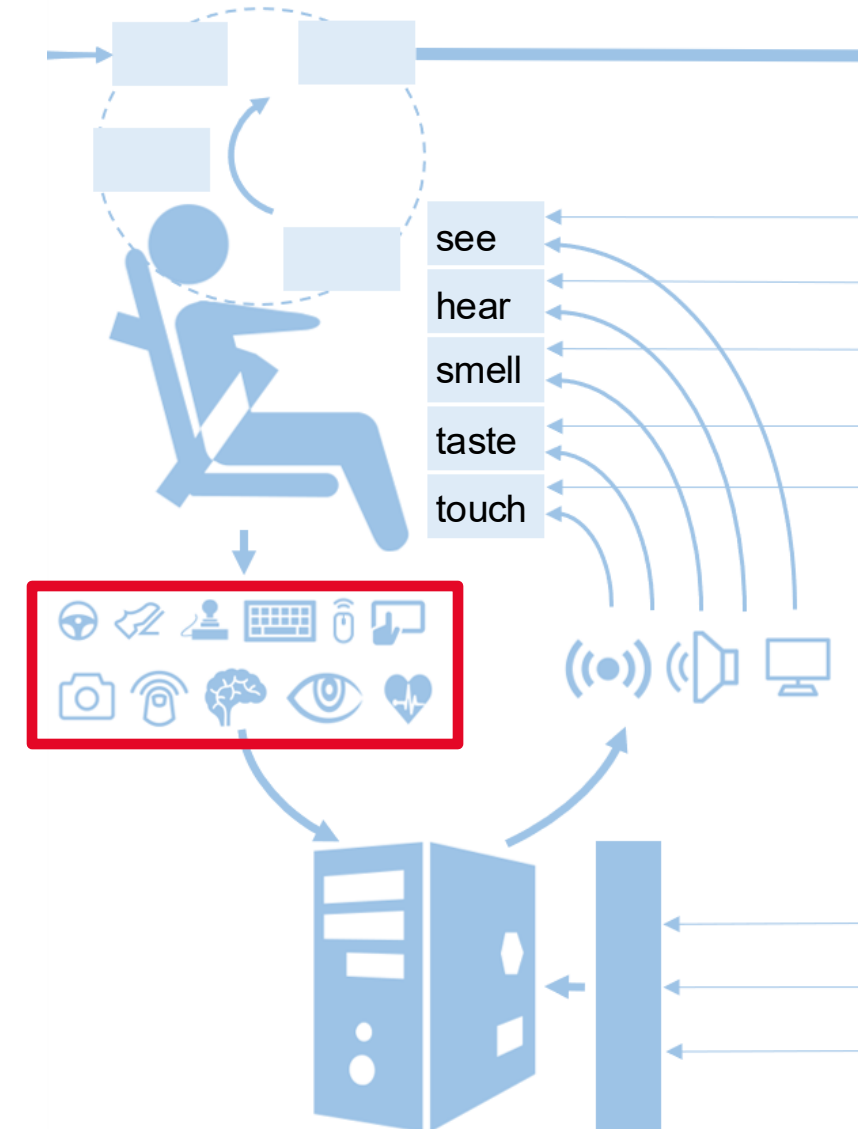


Human Interface Devices

Control Loops

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- Model different processes in a human-machine interaction scenario
- 1) **Output** provided by the machine
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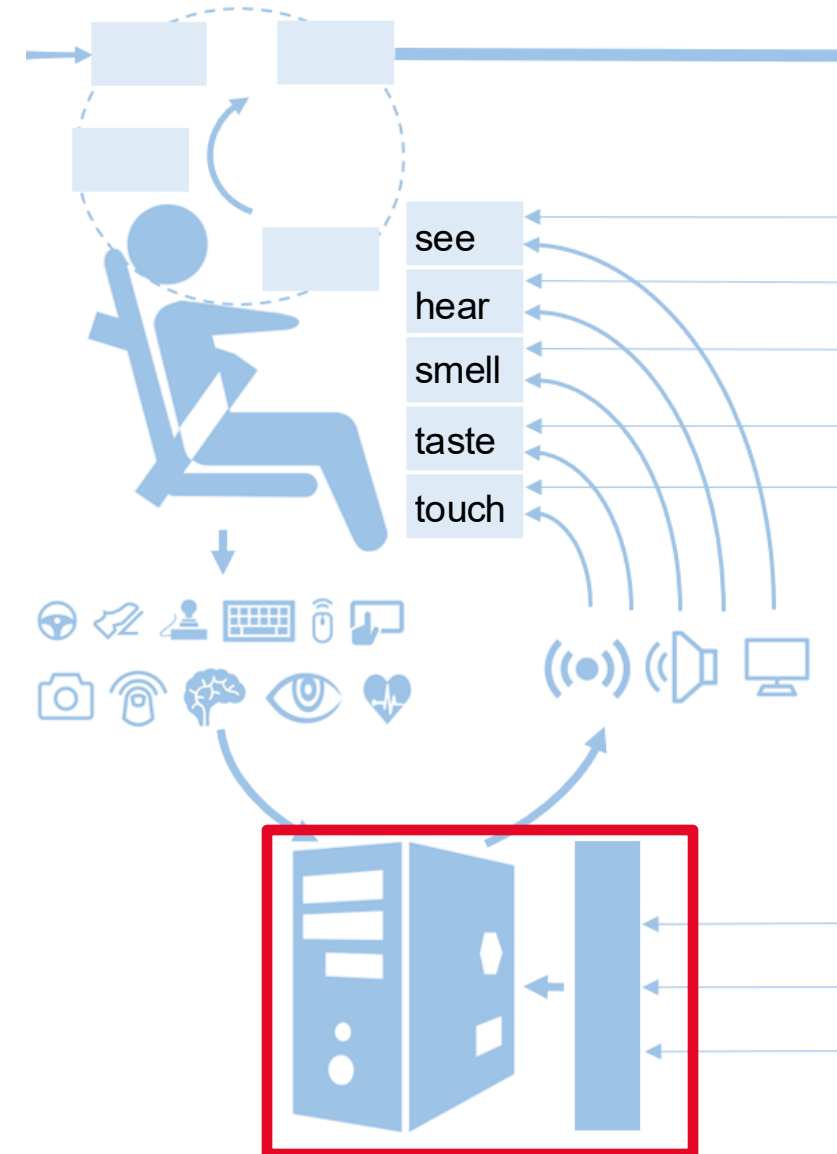


Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

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- 1) **Output** provided by the machine
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- 4) Input by the human
- 5) Computation



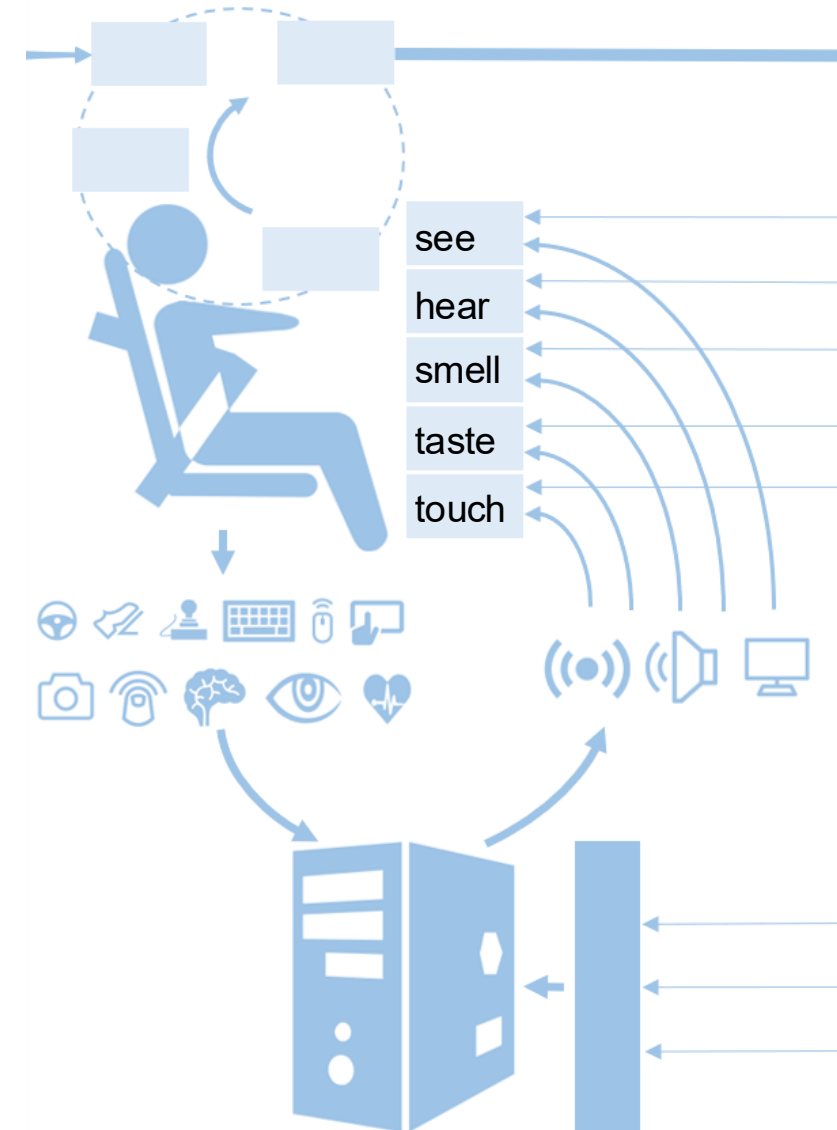
Human Interface Devices

Control Loops

→ Control Loops (in Human-Computer Interaction)

- Model different processes in a human-machine interaction scenario
- 1) **Output** provided by the machine
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→ By **HMI Design**, we can modify parts in the loop and evaluate changes in utility/performance



Human Interface Devices

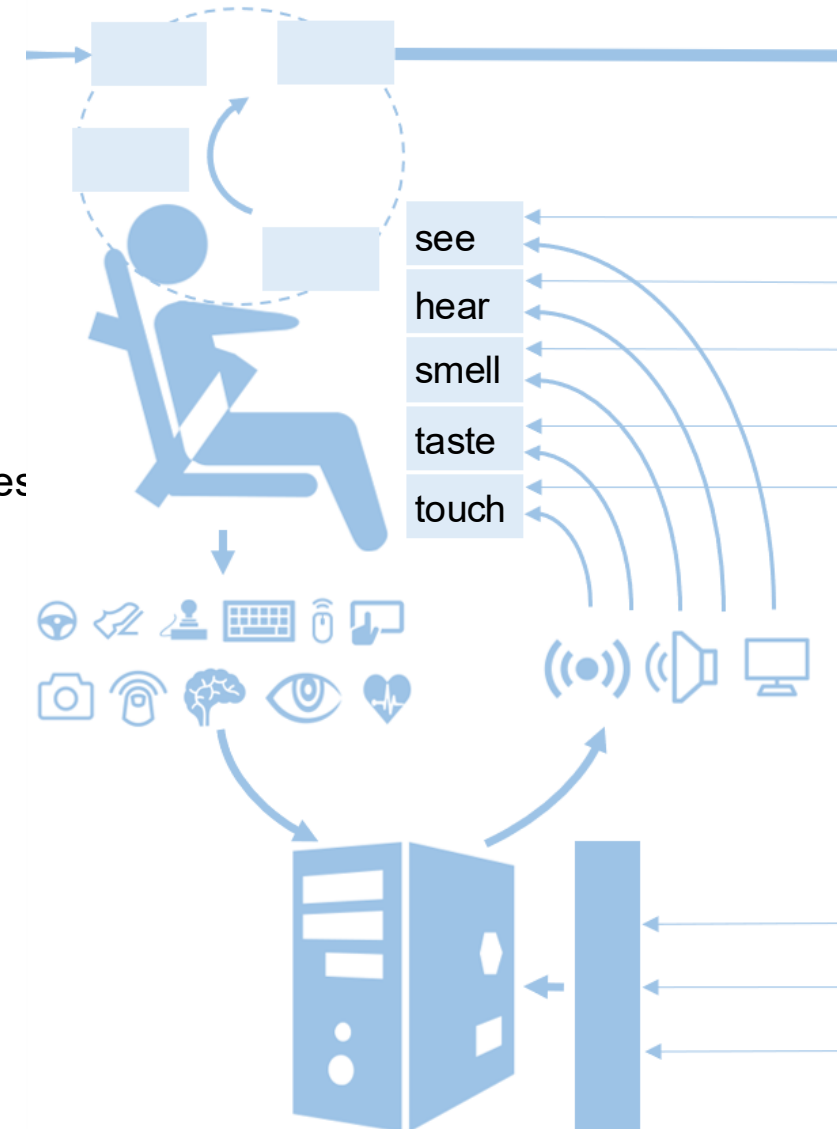
Control Loops

→ Utility/Performance

- There are different optimizations possible
 - Interaction Performance
 - User Experience/Acceptance

→ Simple Example

- Reaction time test – a test subject must press a button as soon as he/she perceives
- How long does this take?
- $RT \text{ (reaction time)} = \text{perception time} + \text{response action time}$
- <https://www.humanbenchmark.com/tests/reactiontime>



Human Interface Devices

Control Loops

Traditional Interface Perspective:
The user sees the real world and
some machine interface

Traditional Interface Perspective:
Make the human part of the loop as
fast, easy, acceptable, etc. as possible

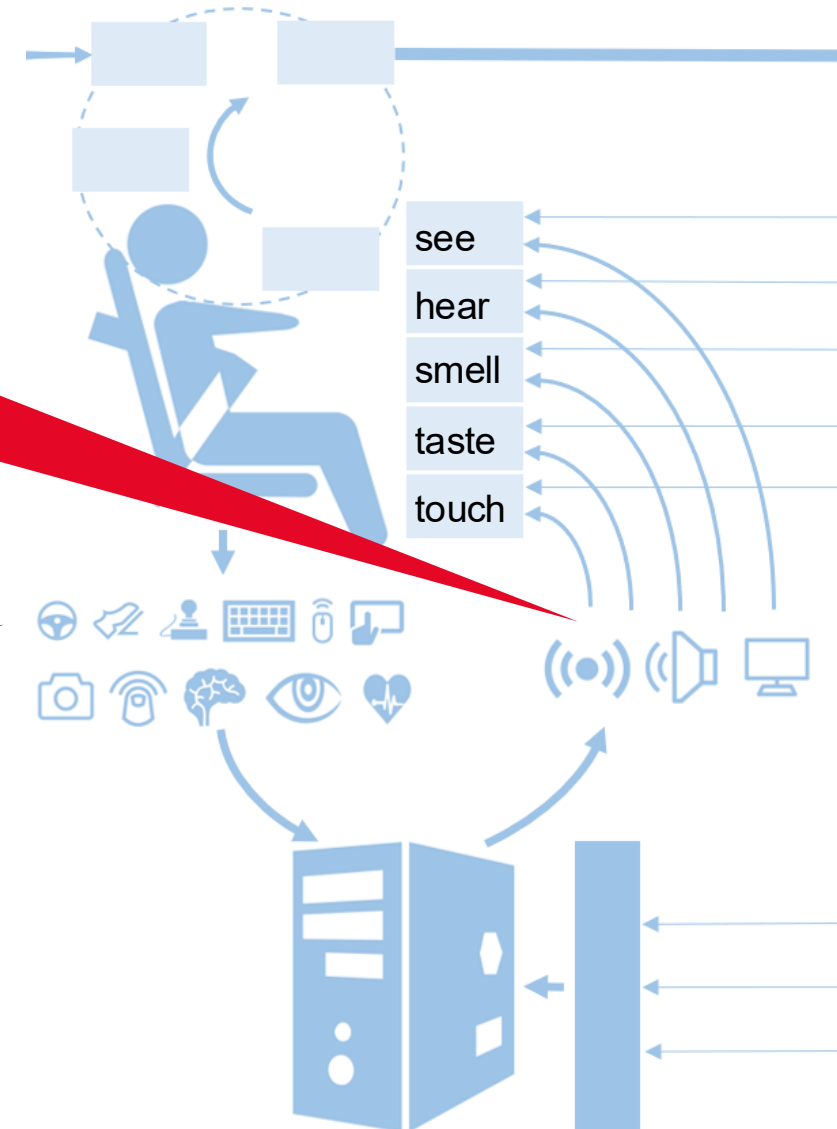


Human Interface Devices

Control Loops

Game Interface Perspective:
There is no “real world” (except for MR),
the virtual world is now a controllable
element (and thus, part of the interface)!

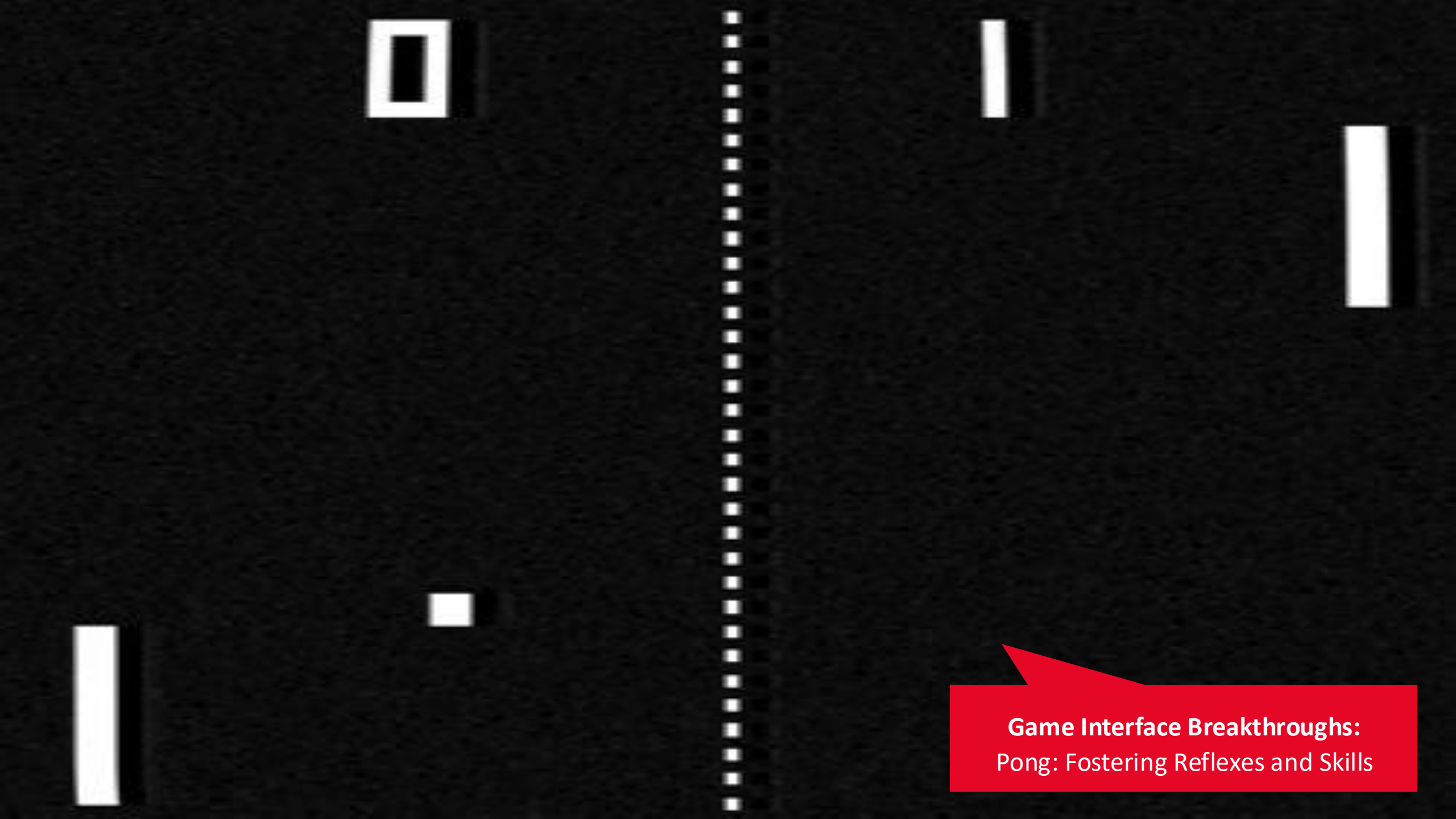
Game Interface Perspective:
Make the human part of the loop as
FUN as possible



A game interface “is the part of the game that allows the user to interact with the game. [It] is the connection between the user and the game”

“the infinitely thin membrane that separates the player
and the game”

It involves input & feedback!



Game Interface Breakthroughs:
Pong: Fostering Reflexes and Skills

1UP
220

HIGH SCORE
1000

2UP
290



Game Interface Breakthroughs:
PacMan: Scores and Puzzle-Solving Activities

Introduction

Somewhere nearby is colossal cave, where others have found fortunes in treasure and gold, though it is rumoured that some who enter are never seen again. Magic is said to work in the cave. I will be your eyes and hands. Direct me with commands of 1 or 2 words. I should warn you that I look at only the first four letters of each word, so you'll have to enter "NORTHEAST" as "NE" to distinguish it from "NORTH". (Should you get stuck, type "HELP" for some general hints. For information on how to end your adventure, etc., type "INFO".)

This program was originally developed by Willie Crowther. Most of the features of the current program were added by Don Woods. The current version was done by Bob Supnik. This version was implemented on the IBM-PC (and compatibles) by Kevin Black.

For further information consult your scroll (READ.ME).

GOOD LUCK!

You are standing at the end of a road be
Around you is a forest. A small stream
down a gully. In the distance there is a

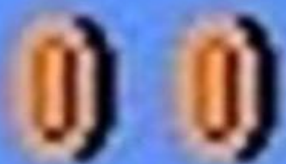
Game Interface Breakthroughs:
Colossal Cave Adventure: Integration of Storylines

MARIO
035350

● × 18

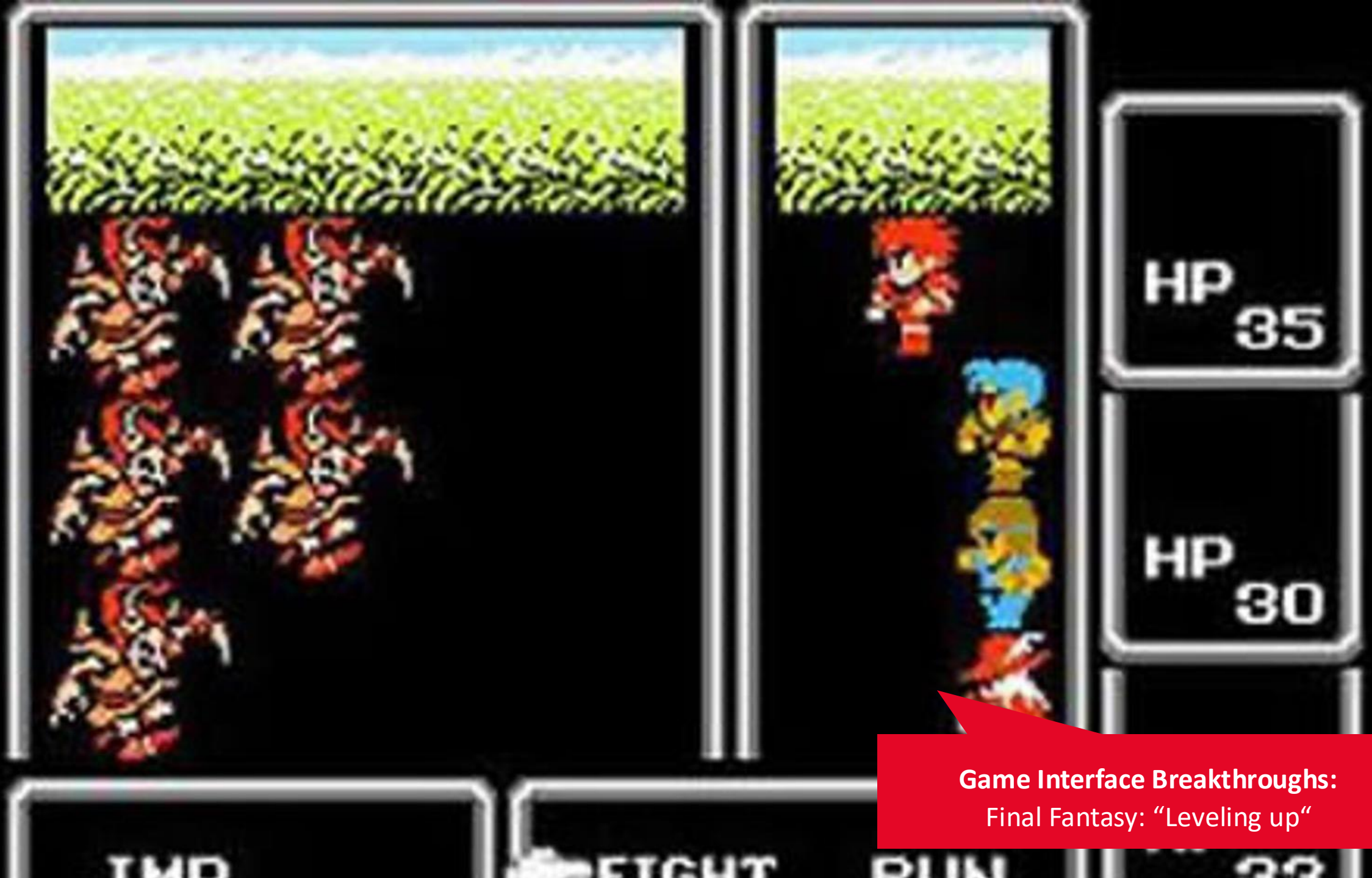
WORLD
1-3

TIME
244



Game Interface Breakthroughs:

Super Mario Bros: Side-Scrolling instead of Page-Flipping



Game Interface Breakthroughs:
Final Fantasy: "Leveling up"



Game Interface Breakthroughs:
Tomb Raider: Action adventures; from a game to a movie



Game Interface Breakthroughs:
Grand Theft Auto: Non-linear gameplay and storylines



Game Interface Breakthroughs:
Crysis: Physics as game element



Game Interface Breakthroughs:
Guitar Freaks: Next generation embodiment



Game Controller Breakthroughs:
Kinect Adventures: Full body movement



Game Controller Breakthroughs:
Missile: Joysticks

Human Interface Devices

Game Interface



Game Controller Breakthroughs:
Nintendo NES Gamepad Controller



Human Interface Devices

Game Interface



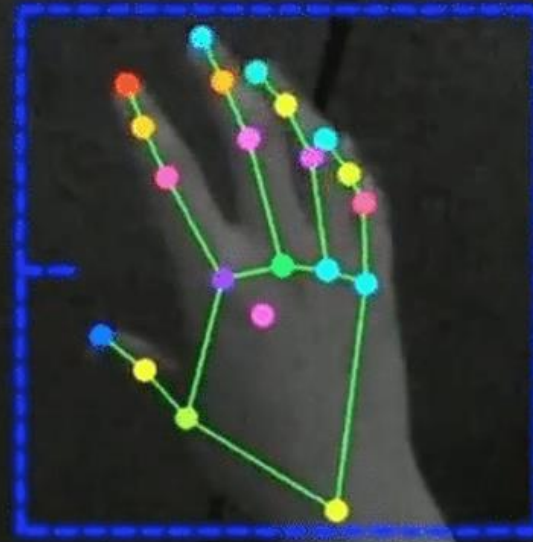
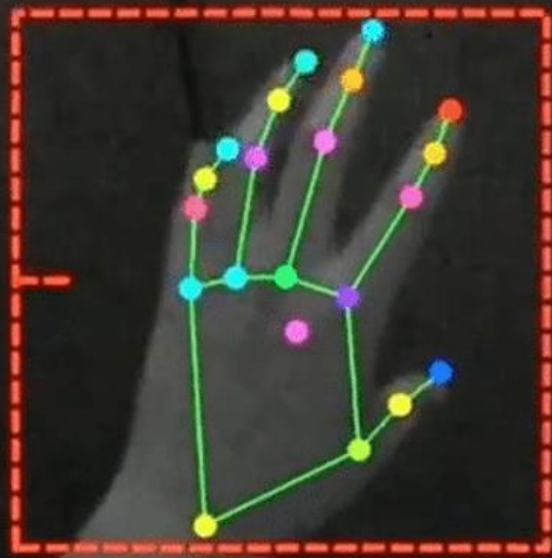
Game Controller Breakthroughs:
Nintendo 64 Analog Stick

Human Interface Devices

Game Interface



Game Controller Breakthroughs:
Nintendo Wii Nunchuck



Game Controller Breakthroughs:
VR Hand Tracking

Game Engine Concepts



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